

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

eTable 1. Demographic and Clinicopathologic Characteristics Among Females Diagnosed With Breast Cancer by Asian American, Native Hawaiian, Pacific Islander Detailed Ethnicity, 2018-2022, SEER-21

Characteristic	Asian American N (%)	Asian Indian and Pakistani N (%)	Chinese N (%)	Filipina N (%)	Japanese N (%)	Korean N (%)	Laotian and Kampuchean N (%)	Vietnamese N (%)	Other Asian American, Asian not otherwise specified (NOS) N (%)	Native Hawaiian, Pacific Islander N (%)	Native Hawaiian N (%)	Guamanian, Chamorro, and Samoan N (%)	Other Pacific Islander and Pacific Islander, NOS N (%)
Age at diagnosis (2000-2022)	138,808	23,402	33,227	34,985	15,772	10,065	1,166	8,909	11,282	9,800	6,546	1,046	2,208
<40 years	10,461 (7.5)	2,549 (10.9)	2,449 (7.4)	1,891 (5.4)	641 (4.1)	791 (7.9)	153 (13.1)	797 (8.9)	1,190 (10.5)	656 (6.7)	398 (6.1)	69 (6.6)	189 (8.6)
40-49 years	31,356 (22.6)	5,768 (24.6)	7,903 (23.8)	6,575 (18.8)	2,615 (16.6)	2,784 (27.7)	311 (26.7)	2,474 (27.8)	2,926 (25.9)	1,761 (18.0)	1,073 (16.4)	201 (19.2)	487 (22.1)
50-59 years	36,259 (26.1)	5,960 (25.5)	9,008 (27.1)	9,471 (27.1)	3,224 (20.4)	2,689 (26.7)	334 (28.6)	2,430 (27.3)	3,143 (27.9)	2,621 (26.7)	1,667 (25.5)	310 (29.6)	644 (29.2)
60-69 years	32,727 (23.6)	5,357 (22.9)	7,214 (21.7)	9,770 (27.9)	3,538 (22.4)	2,191 (21.8)	243 (20.8)	1,994 (22.4)	2,420 (21.5)	2,673 (27.3)	1,867 (28.5)	264 (25.2)	542 (24.5)
70-79 years	19,525 (14.1)	2,931 (12.5)	4,328 (13)	5,455 (15.6)	3,364 (21.3)	1,200 (11.9)	93 (8.0)	914 (10.3)	1,240 (11)	1,551 (15.8)	1,140 (17.4)	156 (14.9)	255 (11.5)
80+ years	8,480 (6.1)	837 (3.6)	2,325 (7)	1,823 (5.2)	2,390 (15.2)	410 (4.1)	32 (2.7)	300 (3.4)	363 (3.2)	538 (5.5)	401 (6.1)	46 (4.4)	91 (4.1)
0-49 years	41,817 (30.1)	8,317 (35.5)	10,352 (31.2)	8,466 (24.2)	3,256 (20.6)	3,575 (35.5)	464 (39.8)	3,271 (36.7)	4,116 (36.5)	2,417 (24.7)	1,471 (22.5)	270 (25.8)	676 (30.6)
50+ years	96,991 (69.9)	15,085 (64.5)	22,875 (68.8)	26,519 (75.8)	12,516 (79.4)	6,490 (64.5)	702 (60.2)	5,638 (63.3)	7,166 (63.5)	7,383 (75.3)	5,075 (77.5)	776 (74.2)	1,532 (69.4)
State (2000-2022)	138,808	23,402	33,227	34,985	15,772	10,065	1,166	8,909	11,282	9,800	6,546	1,046	2,208
California	73,599 (53.0)	6,422 (27.4)	19,099 (57.5)	23,198 (66.3)	7,058 (44.8)	4,970 (49.4)	849 (72.8)	5,776 (64.8)	6,227 (55.2)	2,868 (29.3)	1,256 (19.2)	525 (50.2)	1,087 (49.2)
Connecticut	1,342 (1.0)	451 (1.9)	321 (1.0)	183 (0.5)	56 (0.4)	78 (0.8)	31 (2.7)	87 (1.0)	135 (1.2)	34 (0.3)	13 (0.2)	<5	19 (0.9)
Georgia	3,276 (2.4)	1,236 (5.3)	456 (1.4)	223 (0.6)	132 (0.8)	561 (5.6)	39 (3.3)	319 (3.6)	310 (2.7)	136 (1.4)	80 (1.2)	7 (0.7)	49 (2.2)
Hawaii	12,431 (9.0)	41 (0.2)	1,386 (4.2)	3,546 (10.1)	6,637 (42.1)	566 (5.6)	23 (2.0)	122 (1.4)	110 (1.0)	5,366 (54.8)	4,739 (72.4)	357 (34.1)	270 (12.2)
Idaho	239 (0.2)	19 (0.1)	25 (0.1)	47 (0.1)	49 (0.3)	22 (0.2)	<5	12 (0.1)	65 (0.6)	32 (0.3)	20 (0.3)	<5	11 (0.5)
Illinois	7,365 (5.3)	2381 (10.2)	1,055 (3.2)	1,920 (5.5)	347 (2.2)	647 (6.4)	34 (2.9)	207 (2.3)	774 (6.9)	145 (1.5)	56 (0.9)	7 (0.7)	82 (3.7)
Iowa	392 (0.3)	66 (0.3)	67 (0.2)	62 (0.2)	13 (0.1)	30 (0.3)	20 (1.7)	59 (0.7)	75 (0.7)	36 (0.4)	14 (0.2)	<5	21 (1.0)
Kentucky	421 (0.3)	94 (0.4)	52 (0.2)	59 (0.2)	44 (0.3)	51 (0.5)	<5	34 (0.4)	85 (0.8)	34 (0.3)	21 (0.3)	<5	9 (0.4)
Louisiana	574 (0.4)	134 (0.6)	59 (0.2)	67 (0.2)	27 (0.2)	36 (0.4)	7 (0.6)	166 (1.9)	78 (0.7)	33 (0.3)	10 (0.2)	<5	19 (0.9)
New Jersey	9,008 (6.5)	3361 (14.4)	1,716 (5.2)	2,130 (6.1)	248 (1.6)	1,059 (10.5)	16 (1.4)	179 (2.0)	299 (2.7)	181 (1.8)	74 (1.1)	28 (2.7)	79 (3.6)
New Mexico	342 (0.2)	50 (0.2)	48 (0.1)	55 (0.2)	40 (0.3)	33 (0.3)	<5	23 (0.3)	91 (0.8)	30 (0.3)	18 (0.3)	<5	8 (0.4)
New York	18,691 (13.5)	5,773 (24.7)	7,110 (21.4)	2,199 (6.3)	646 (4.1)	1,361 (13.5)	67 (5.7)	305 (3.4)	1230 (10.9)	227 (2.3)	56 (0.9)	24 (2.3)	147 (6.7)
Texas	10,547 (7.6)	3,333 (14.2)	1,732 (5.2)	1,227 (3.5)	332 (2.1)	602 (6)	63 (5.4)	1,567 (17.6)	1691 (15.0)	421 (4.3)	118 (1.8)	34 (3.3)	269 (12.2)
Utah	581 (0.4)	41 (0.2)	101 (0.3)	69 (0.2)	143 (0.9)	49 (0.5)	13 (1.1)	53 (0.6)	112 (1.0)	257 (2.6)	71 (1.1)	48 (4.6)	138 (6.3)
Stage (2004-2022)	124,828	21,730	29,919	31,106	13,281	9,104	1,058	8,193	10,437	8,733	5,753	930	2,050
Localized	80,003 (64.1)	12,951 (59.6)	19,872 (66.4)	19,565 (62.9)	9,452 (71.2)	5,815 (63.9)	571 (54)	5,138 (62.7)	6,639 (63.6)	5,280 (60.5)	3,661 (63.6)	470 (50.5)	1,149 (56.0)
Regional	35,418 (28.4)	6,791 (31.3)	8,027 (26.8)	9,104 (29.3)	3,056 (23)	2,694 (29.6)	365 (34.5)	2,404 (29.3)	2,977 (28.5)	2,606 (29.8)	1,640 (28.5)	328 (35.3)	638 (31.1)
Distant	6,133 (4.9)	1,312 (6.0)	1,221 (4.1)	1,759 (5.7)	501 (3.8)	357 (3.9)	89 (8.4)	375 (4.6)	519 (5.0)	625 (7.2)	335 (5.8)	99 (10.6)	191 (9.3)
Unknown/unstaged	3,274 (2.6)	676 (3.1)	799 (2.7)	678 (2.2)	272 (2)	238 (2.6)	33 (3.1)	276 (3.4)	302 (2.9)	222 (2.5)	117 (2.0)	33 (3.5)	72 (3.5)
Subtype (2010-2022)	97,283	18,072	23,148	23,602	9,489	7,068	837	6,504	8,563	6,731	4,406	702	1,623
HR+/ ERBB2-	64,514 (66.3)	11,751 (65)	15,156 (65.5)	15,748 (66.7)	7,060 (74.4)	4,652 (65.8)	461 (55.1)	3,990 (61.3)	5,696 (66.5)	4,804 (71.4)	3,279 (74.4)	476 (67.8)	1,049 (64.6)
HR-/ ERBB2+	5,659 (5.8)	947 (5.2)	1,423 (6.1)	1,533 (6.5)	338 (3.6)	392 (5.5)	78 (9.3)	463 (7.1)	485 (5.7)	331 (4.9)	199 (4.5)	26 (3.7)	106 (6.5)
HR+/ ERBB2+	10,745 (11.0)	1,995 (11.0)	2,526 (10.9)	2,882 (12.2)	667 (7.0)	764 (10.8)	101 (12.1)	828 (12.7)	982 (11.5)	718 (10.7)	437 (9.9)	91 (13.0)	190 (11.7)
HR-/ ERBB2- (TNBC)	8,613 (8.9)	1,958 (10.8)	2,011 (8.7)	1,712 (7.3)	866 (9.1)	714 (10.1)	105 (12.5)	562 (8.6)	685 (8.0)	485 (7.2)	299 (6.8)	55 (7.8)	131 (8.1)
Unknown	7,752 (8)	1,421 (7.9)	2,032 (8.8)	1,727 (7.3)	558 (5.9)	546 (7.7)	92 (11.0)	661 (10.2)	715 (8.3)	393 (5.8)	192 (4.4)	54 (7.7)	147 (9.1)

eTable 2. Demographic and Clinicopathologic Characteristics Among Females Diagnosed With Breast Cancer by Aggregated Racial and Ethnic Groups, 2000-2022, SEER-21

Characteristic	Asian American, Native Hawaiian, and Pacific Islander N (%)	Hispanic N (%)	Non-Hispanic American Indian/Alaska Native N (%)	Non-Hispanic Black N (%)	Non-Hispanic White N (%)
Age at diagnosis (2000-2022)	148,612	277,063	7,181	252,417	1,530,634
<40 years	11,117 (7.5)	24,616 (8.9)	468 (6.5)	18,253 (7.2)	57,982 (3.8)
40-49 years	33,117 (22.3)	61,151 (22.1)	1,295 (18.0)	45,372 (18.0)	205,119 (13.4)
50-59 years	3,8881 (26.2)	70,911 (25.6)	1,846 (25.7)	64,108 (25.4)	332,091 (21.7)
60-69 years	35,401 (23.8)	63,392 (22.9)	1,897 (26.4)	62,255 (24.7)	398,785 (26.1)
70-79 years	21,077 (14.2)	3,9061 (14.1)	1,182 (16.5)	41,076 (16.3)	329,777 (21.5)
80+ years	9,019 (6.1)	17,932 (6.5)	493 (6.9)	21,353 (8.5)	206,880 (13.5)
0-49 years	44,234 (29.8)	85,767 (31.0)	1,763 (24.6)	63,625 (25.2)	263,101 (17.2)
50+ years	104,378 (70.2)	191,296 (69.0)	5,418 (75.4)	188,792 (74.8)	1,267,533 (82.8)
State (2000-2022)	148,612	277,063	7,181	252,417	1,530,634
California	76,470 (51.5)	107,282 (38.7)	2,823 (39.3)	36,783 (14.6)	350,056 (22.9)
Connecticut	1,377 (0.9)	5,125 (1.8)	86 (1.2)	5,312 (2.1)	60,207 (3.9)
Georgia	3,412 (2.3)	5,046 (1.8)	184 (2.6)	43,398 (17.2)	103,126 (6.7)
Hawaii	17,797 (12.0)	731 (0.3)	104 (1.4)	203 (0.1)	5,841 (0.4)
Idaho	271 (0.2)	1,019 (0.4)	253 (3.5)	48 (0.02)	23,433 (1.5)
Illinois	7,510 (5.1)	13,628 (4.9)	211 (2.9)	30,060 (11.9)	169,448 (11.1)
Iowa	428 (0.3)	661 (0.2)	113 (1.6)	855 (0.3)	52,905 (3.5)
Kentucky	455 (0.3)	681 (0.2)	35 (0.5)	5,065 (2.0)	68,665 (4.5)
Louisiana	607 (0.4)	1,467 (0.5)	134 (1.9)	22,398 (8.9)	50,383 (3.3)
New Jersey	9,189 (6.2)	15,458 (5.6)	116 (1.6)	19,226 (7.6)	123,220 (8.1)
New Mexico	372 (0.3)	10,360 (3.7)	1,320 (18.4)	417 (0.2)	18,511 (1.2)
New York	18,918 (12.7)	36,763 (13.3)	523 (7.3)	46,324 (18.4)	252,014 (16.5)
Texas	10,968 (7.4)	76,405 (27.6)	1,121 (15.6)	42,174 (16.7)	224,204 (14.6)
Utah	838 (0.6)	2,437 (0.9)	158 (2.2)	154 (0.1)	28,621 (1.9)
Stage (2004-2022)	133,564	246,480	6,446	218,942	1,276,033
Localized	85,285 (63.9)	140,838 (57.1)	3,870 (60.0)	120,239 (54.9)	835,460 (65.5)
Regional	38,025 (28.5)	81,005 (32.9)	1,933 (30.0)	71,894 (32.8)	331,708 (26.0)
Distant	6,758 (5.1)	15,028 (6.1)	393 (6.1)	19,147 (8.7)	69,757 (5.5)
Unknown/unstaged	3,496 (2.6)	9,609 (3.9)	250 (3.9)	7,662 (3.5)	39,108 (3.1)
Subtype (2010-2022)	104,015	188,638	5,022	160,981	895,840
HR+/ERBB2-	69,318 (66.6)	118,098 (62.6)	3,261 (64.9)	91,210 (56.7)	631,802 (70.5)
HR-/ ERBB2+	5,990 (5.8)	9,368 (5.0)	237 (4.7)	7,928 (4.9)	30,578 (3.4)
HR+/ ERBB2+	11,463 (11.0)	20,285 (10.8)	509 (10.1)	15,929 (9.9)	78,902 (8.8)
HR-/ ERBB2- (TNBC)	9,099 (8.7)	21,737 (11.5)	502 (10.0)	30,529 (19.0)	79,003 (8.8)
Unknown	8,145 (7.8)	19,150 (10.2)	513 (10.2)	15,385 (9.6)	75,555 (8.4)

eTable 3. Age-Adjusted Incidence Rates and 95% CIs for Females Diagnosed With Breast Cancer by Aggregated Racial and Ethnic Group, 2018-2022, SEER-21

Characteristic	Asian American, Native Hawaiian, andPacific Islander	Hispanic	Non-Hispanic American Indian/Alaska Native	Non-Hispanic Black	Non-Hispanic White
Overall	112.9 (111.8, 113.9)	104.0 (103.3, 104.7)	98.7 (94.5, 103.1)	131.2 (130.1, 132.2)	139.5 (139.0, 140.0)
Age at diagnosis					
<40 years	13.9 (13.5, 14.4)	12.8 (12.5, 13.1)	12.2 (10.1, 14.6)	17.0 (16.5, 17.6)	15.1 (14.9, 15.4)
40-49 years	178.4 (174.9, 181.9)	134.7 (132.7, 136.7)	142.6 (127.8, 158.7)	162.0 (158.8, 165.2)	179.6 (177.8, 181.4)
50-59 years	245.8 (241.3, 250.3)	207.4 (204.6, 210.2)	204.2 (187.0, 222.6)	257.5 (253.5, 261.5)	265.6 (263.6, 267.7)
60-69 years	310.0 (304.5, 315.5)	310.7 (306.6, 314.9)	277.6 (256.5, 300.0)	378.2 (372.9, 383.6)	397.3 (394.8, 399.7)
70-79 years	331.1 (323.7, 338.5)	352.8 (346.8, 359.0)	325.0 (294.5, 357.8)	452.8 (444.9, 460.9)	511.4 (508, 514.7)
80+ years	240.5 (232.1, 249.1)	284.6 (277.1, 292.2)	231.1 (195.7, 271.1)	408.8 (398.3, 419.5)	438.3 (434.2, 442.5)
0-49 years	48.9 (48.1, 49.8)	38.7 (38.2, 39.2)	39.9 (36.4, 43.8)	47.9 (47.1, 48.7)	50.1 (49.7, 50.6)
50+ years	280.2 (277.3, 283.2)	275.0 (272.7, 277.2)	252.5 (240.6, 264.9)	349.2 (346.2, 352.2)	373.5 (372.2, 374.9)
Stage					
Localized	73.0 (72.2, 73.9)	62.0 (61.5, 62.6)	59.7 (56.5, 63.1)	75.2 (74.4, 76.0)	93.6 (93.2, 94.0)
Regional	30.8 (30.3, 31.4)	31.4 (31, 31.8)	28.2 (25.9, 30.6)	40.1 (39.6, 40.7)	34.8 (34.5, 35.0)
Distant	6.2 (5.9, 6.4)	6.7 (6.5, 6.8)	6.6 (5.6, 7.8)	11.9 (11.6, 12.2)	7.8 (7.7, 7.9)
Subtype					
HR+/ ERBB2-	77.3 (76.4, 78.1)	68.6 (68.0, 69.1)	65.5 (62.1, 69.0)	77.9 (77.1, 78.6)	101.6 (101.2, 102.0)
HR-/ ERBB2+	6.3 (6.1, 6.6)	4.8 (4.7, 5.0)	4.0 (3.2, 5.0)	6.2 (5.9, 6.4)	4.7 (4.7, 4.8)
HR+/ ERBB2+	12.0 (11.7, 12.4)	10.4 (10.2, 10.6)	10.6 (9.2, 12.1)	12.6 (12.3, 12.9)	12.8 (12.7, 13.0)
HR-/ ERBB2- (TNBC)	10.0 (9.7, 10.3)	11.9 (11.7, 12.2)	10.6 (9.2, 12.1)	25.4 (25.0, 25.9)	12.7 (12.5, 12.8)

eTable 4. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Age Group at Diagnosis, 2000-2022, SEER-21

	Segment 1		Segment 2		Segment 3		Segment 4	
	Years	APC 1 (95% CI)	Years	APC 2 (95% CI)	Years	APC 3 (95% CI)	Years	APC 4 (95% CI)
All Ages								
Asian American	2000-2012	0.80 (-0.87, 1.31)	2012-2022	2.34 (1.83, 3.68)				
Asian Indian and Pakistani	2000-2008	0.93 (-3.79, 2.37)	2008-2015	4.17 (3.20, 7.81)	2015-2022	1.94 (0.42, 2.70)		
Chinese	2000-2016	1.36 (-0.20, 1.91)	2016-2022	4.39 (2.58, 7.70)				
Filipina	2000-2004	-0.93 (-4.93, 1.00)	2004-2022	1.26 (1.03, 2.11)				
Japanese	2000-2006	-0.97 (-8.28, 1.62)	2006-2022	1.97 (1.11, 5.15)				
Korean	2000-2010	2.95 (-0.96, 9.19)	2010-2013	-2.33 (-5.33, 6.92)	2013-2019	6.99 (-1.82, 12.33)	2019-2022	1.51 (-1.39, 5.29)
Laotian and Kampuchean	2000-2022	1.59 (0.54, 3.01)						
Vietnamese	2000-2015	1.80 (-3.95, 14.07)	2015-2022	4.12 (2.32, 8.10)				
Native Hawaiian/ Pacific Islander	2000-2022	0.84 (0.48, 1.29)						
Native Hawaiian	2000-2022	0.68 (0.13, 1.38)						
Guamanian, Chamorro, and Samoan	2000-2022	0.88 (-0.52, 2.63)						
Age at Diagnosis < 50 Years								
Asian American	2000-2016	1.19 (-1.16, 1.93)	2016-2022	2.90 (1.57, 5.29)				
Asian Indian and Pakistani	2000-2022	2.09 (1.65, 2.72)						
Chinese	2000-2017	1.16 (-0.56, 1.69)	2017-2022	4.57 (2.03, 8.36)				
Filipina	2000-2022	0.40 (-0.01, 0.86)						
Japanese	2000-2022	2.41 (1.77, 3.14)						
Korean	2000-2022	2.86 (2.25, 3.62)						
Laotian and Kampuchean	2002-2022	1.67 (-0.06, 3.76) ^						
Vietnamese	2000-2016	1.30 (-3.32, 2.90)	2016-2022	4.30 (1.98, 8.93)				
Native Hawaiian/ Pacific Islander	2000-2022	0.99 (0.39, 1.70)						
Native Hawaiian	2000-2022	1.14 (0.38, 2.02)						
Guamanian, Chamorro, and Samoan	2002-2022	-1.79 (-3.64, 0.07) ^						
Age at Diagnosis ≥ 50 Years								
Asian American	2000-2003	-2.09 (-5.31, 0.97)	2003-2014	1.27 (0.23, 3.32)	2014-2022	2.48 (1.82, 3.88)		
Asian Indian and Pakistani	2000-2004	-0.56 (-7.22, 3.13)	2004-2022	3.15 (2.79, 4.40)				

Chinese	2000-2014	1.24 (-0.29, 1.94)	2014-2022	3.97 (2.90, 6.59)				
Filipina	2000-2004	-1.14 (-5.9, 1.18)	2004-2022	1.56 (1.33, 2.36)				
Japanese	2000-2006	-2.49 (-11.18, 0.96)	2006-2022	1.79 (0.86, 5.35)				
Korean	2000-2014	2.01 (-0.80, 2.98)	2014-2022	5.45 (4.00, 9.03)				
Laotian and Kampuchean	2002-2022	2.50 (1.69, 3.6) ^						
Vietnamese	2000-2022	2.87 (2.37, 3.58)						
Native Hawaiian/ Pacific Islander	2000-2022	0.77 (0.31, 1.35)						
Native Hawaiian	2000-2022	0.53 (-0.07, 1.27)						
Guamanian, Chamorro, and Samoan	2002-2012	5.18 (2.64, 12.59) ^	2011-2022	-1.01 (-5.27, 1.21) ^				

Abbreviations: AANHPI (Asian American, Native Hawaiian, Pacific Islander); APC (Annual Percent Change); CI (Confidence Interval)

^ Tri-annual percent change (2002-2004, 2005-2007, 2008-2010, 2011-2013, 2014-2016, 2017-2019, 2020-2022) *excl 2000-2001

eTable 5. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Summary Stage at Diagnosis, 2004-2022, SEER-21

	Segment 1		Segment 2		Segment 3		Segment 4	
	Years	APC 1 (95% CI)	Years	APC 2 (95% CI)	Years	APC 3 (95% CI)	Years	APC 4 (95% CI)
Localized								
Asian American	2004-2016	1.84 (-0.01, 2.97)	2016-2022	2.86 (2.13, 4.25)				
Asian Indian and Pakistani	2004-2009	0.11 (-6.97, 2.71)	2009-2012	10.31 (5.52, 13.1)	2012-2022	3.11 (2.16, 3.72)		
Chinese	2004-2016	1.77 (-1.71, 2.56)	2016-2022	4.41 (2.71, 7.57)				
Filipina	2004-2019	1.98 (1.71, 3.86)	2019-2022	-0.94 (-2.92, 1.49)				
Japanese	2004-2022	2.22 (1.71, 2.75)						
Korean	2004-2010	4.25 (3.30, 5.94)	2010-2013	-4.02 (-5.81, -0.75)	2013-2019	8.04 (7.28, 10.9)	2019-2022	3.04 (1.11, 5.25)
Laotian and Kampuchean	2005-2022	2.35 (-0.95, 6.63) ^						
Vietnamese	2004-2022	2.96 (2.20, 3.95)						
Native Hawaiian/ Pacific Islander	2004-2022	1.15 (0.45, 1.98)						
Native Hawaiian	2004-2022	1.38 (0.43, 2.48)						
Guamanian, Chamorro, and Samoan	2005-2022	-0.17 (-3.07, 3.25) ^						
Regional								
Asian American	2004-2016	0.15 (-4.55, 6.11)	2016-2022	2.53 (0.63, 6.15)				
Asian Indian and Pakistani	2004-2022	1.34 (0.62, 2.27)						
Chinese	2004-2014	-0.34 (-6.21, 1.02)	2014-2022	3.23 (1.45, 7.99)				
Filipina	2004-2022	0.20 (-0.34, 0.77)						
Japanese	2004-2022	0.70 (-0.32, 1.71)						
Korean	2004-2022	2.12 (1.26, 3.09)						
Laotian and Kampuchean	2005-2022	2.90 (-1.13, 8.33) ^						
Vietnamese	2004-2009	-6.32 (-19.68, 0.09)	2009-2022	3.37 (2.03, 7.49)				
Native Hawaiian/ Pacific Islander	2004-2022	0.11 (-0.55, 0.85)						
Native Hawaiian	2004-2022	-0.02 (-0.81, 0.83)						
Guamanian, Chamorro, and Samoan	2005-2022	0.17 (-2.49, 3.13) ^						
Distant								
Asian American	2004-2022	3.11 (2.48, 3.91)						

Asian Indian and Pakistani	2004-2022	4.02 (2.74, 5.72)						
Chinese	2004-2022	4.52 (2.55, 7.09)						
Filipina	2004-2022	2.33 (1.45, 3.37)						
Japanese	2005-2022	2.10 (-1.85, 6.26) ^						
Korean	2005-2022	3.95 (0.01, 8.92) ^						
Laotian and Kampuchean	--	--						
Vietnamese	2005-2022	2.73 (-1.15, 8.00) ^						
Native Hawaiian/ Pacific Islander	2004-2022	2.42 (-0.01, 5.31)						
Native Hawaiian	2005-2022	1.11 (-2.25, 5.14) ^						
Guamanian, Chamorro, and Samoan	--	--						

Abbreviations: AANHPI (Asian American, Native Hawaiian, Pacific Islander); APC (Annual Percent Change); CI (Confidence Interval)

^ Tri-annual percent change: 2005-2007, 2008-2010, 2011-2013, 2014-2016, 2017-2019, 2020-2022 * excl 2004

Laotian, Kampuchean and Guamanian, Chamorro, Samoan rates and tri-annual percent change suppressed due to <16 cases for at least one three-year period.

eTable 6. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Breast Cancer Subtype, 2010-2022, SEER-21

	Segment 1		Segment 2		Segment 3	
	Years	APC 1 (95% CI)	Years	APC 2 (95% CI)	Years	APC 3 (95% CI)
HR+/ERBB2-						
Asian American	2010-2022	3.10 (2.59, 3.67)				
Asian Indian and Pakistani	2010-2022	3.61 (3.00, 4.34)				
Chinese	2010-2022	4.04 (3.29, 4.85)				
Filipina	2010-2022	2.15 (1.56, 2.76)				
Japanese	2010-2022	2.10 (0.74, 3.45)				
Korean	2010-2022	6.00 (3.62, 8.78)				
Laotian and Kampuchean	2010-2022	2.03 (-2.59, 7.38)				
Vietnamese	2010-2022	5.01 (3.78, 6.52)				
Native Hawaiian/ Pacific Islander	2010-2022	0.89 (0.00, 1.84)				
Native Hawaiian	2010-2022	0.79 (-0.42, 2.06)				
Guamanian, Chamorro, and Samoan	2010-2022	0.47 (-3.5, 4.72)				
HR-/ ERBB 2+						
Asian American	2010-2022	1.87 (0.71, 3.10)				
Asian Indian and Pakistani	2010-2013	19.43 (6.58, 41.89)	2013-2022	-1.01 (-4.14, 0.81)		
Chinese	2010-2019	0.67 (-9.81, 11.46)	2019-2022	11.47 (2.21, 18.66)		
Filipina	2010-2022	1.89 (-0.14, 3.92)				
Japanese	2010-2022	5.72 (2.22, 9.38)				
Korean	2010-2013	-12.35 (-24.93, -1.87)	2013-2022	2.75 (-0.02, 11.02)		
Laotian and Kampuchean	--	--				
Vietnamese	2010-2022	0.60 (-2.44, 3.87)				
Native Hawaiian/ Pacific Islander	2010-2022	1.91 (-1.43, 5.48)				
Native Hawaiian	2011-2022	5.07 (0.50, 10.81) ^				
Guamanian, Chamorro, and Samoan	--	--				
HR+/ ERBB 2+						
Asian American	2010-2016	5.48 (4.38, 6.90)	2016-2022	-1.36 (-2.43, -0.44)		
Asian Indian and Pakistani	2010-2012	24.97 (5.62, 41.46)	2012-2022	1.55 (-1.62, 3.01)		
Chinese	2010-2017	6.68 (4.88, 9.38)	2017-2022	-3.21 (-6.68, -0.8)		
Filipina	2010-2015	5.24 (1.09, 10.37)	2015-2019	-1.29 (-3.06, 8.22)	2019-2022	-5.97 (-8.79, -3.09)
Japanese	2010-2022	2.91 (0.01, 5.70)				
Korean	2010-2022	2.42 (-0.43, 5.50)				

Laotian and Kampuchean	2011-2022	0.03 (-6.68, 8.00) ^				
Vietnamese	2010-2016	9.73 (0.89, 69.56)	2016-2022	0.12 (-14.23, 7.94)		
Native Hawaiian/ Pacific Islander	2010-2022	1.91 (-1.22, 5.55)				
Native Hawaiian	2010-2022	1.34 (-1.17, 4.02)				
Guamanian, Chamorro, and Samoan	2011-2022	-2.30 (-6.70, 2.45) ^				
HR-/ ERBB 2- (TNBC)						
Asian American	2010-2016	-0.16 (-5.85, 1.75)	2016-2022	4.47 (2.71, 8.52)		
Asian Indian and Pakistani	2010-2022	3.06 (1.91, 4.42)				
Chinese	2010-2017	0.29 (-6.42, 2.21)	2017-2022	6.17 (3.30, 11.73)		
Filipina	2010-2013	7.16 (2.93, 19.81)	2013-2016	-4.77 (-8.16, -0.36)	2016-2022	4.00 (1.93, 9.44)
Japanese	2010-2016	-6.94 (-14.39, -3.62)	2016-2019	25.38 (13.28, 33.80)	2019-2022	-11.21 (-19.03, -2.14)
Korean	2010-2022	2.38 (-0.34, 5.36)				
Laotian and Kampuchean	2011-2022	1.16 (-3.74, 6.82) ^				
Vietnamese	2010-2022	3.86 (1.38, 6.85)				
Native Hawaiian/ Pacific Islander	2010-2022	1.10 (-3.00, 5.53)				
Native Hawaiian	2011-2022	1.37 (-1.83, 5.00) ^				
Guamanian, Chamorro, and Samoan	--	--				

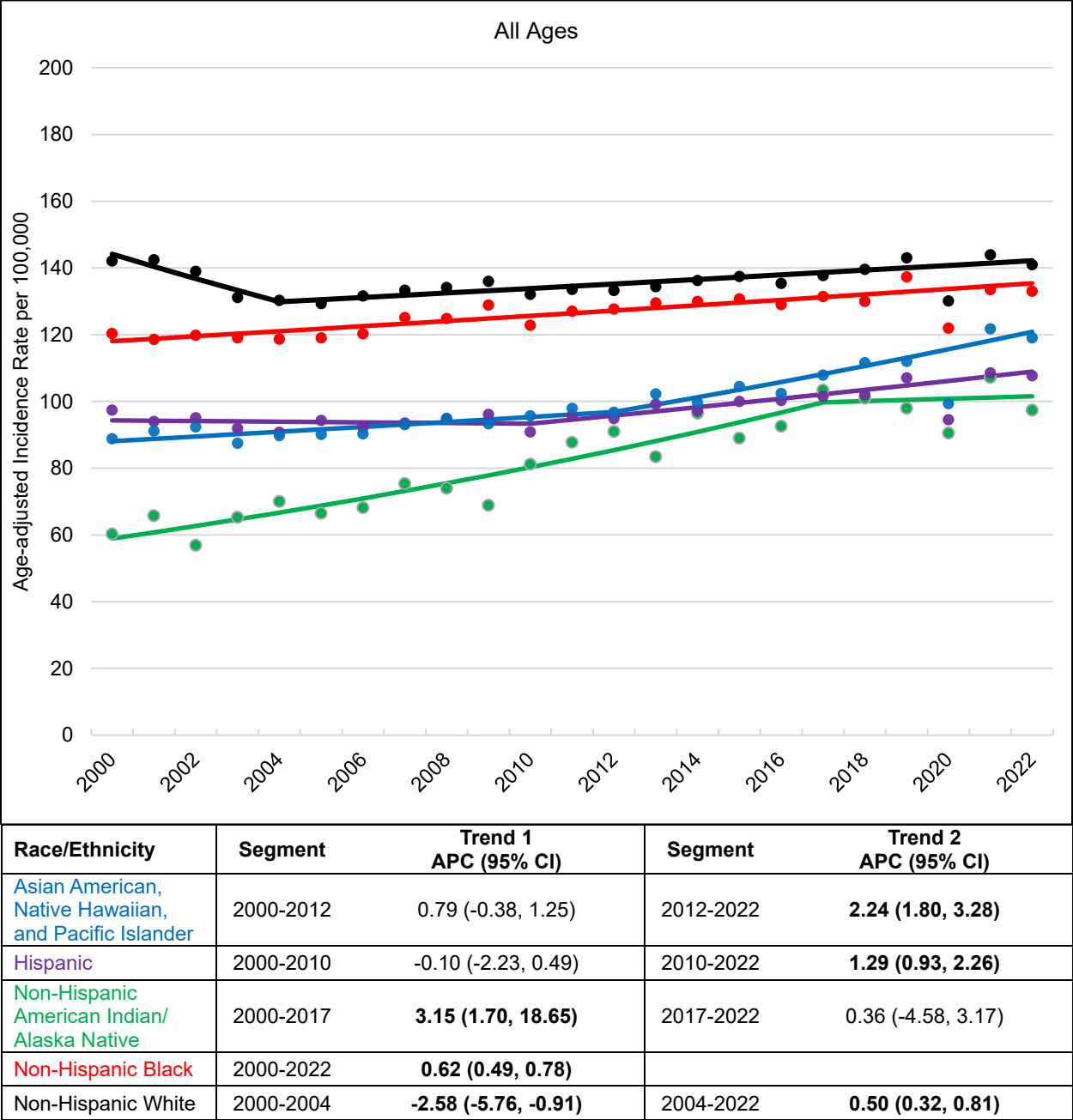
Abbreviations: AANHPI (Asian American, Native Hawaiian, Pacific Islander); APC (Annual Percent Change); CI (Confidence Interval); TNBC (Triple Negative Breast Cancer)

^ Tri-annual percentage change: 2011-2013, 2014-2016, 2017-2019, 2020-2022 *excl 2010

Laotian, Kampuchean and Guamanian, Chamorro, Samoan tri-annual percent change in trend suppressed for HR-/ **ERBB** 2+ due to <16 cases for at least one three-year period.

Guamanian, Chamorro, Samoan tri-annual percent change in trend suppressed for TNBC due to <16 cases for at least one three-year period.

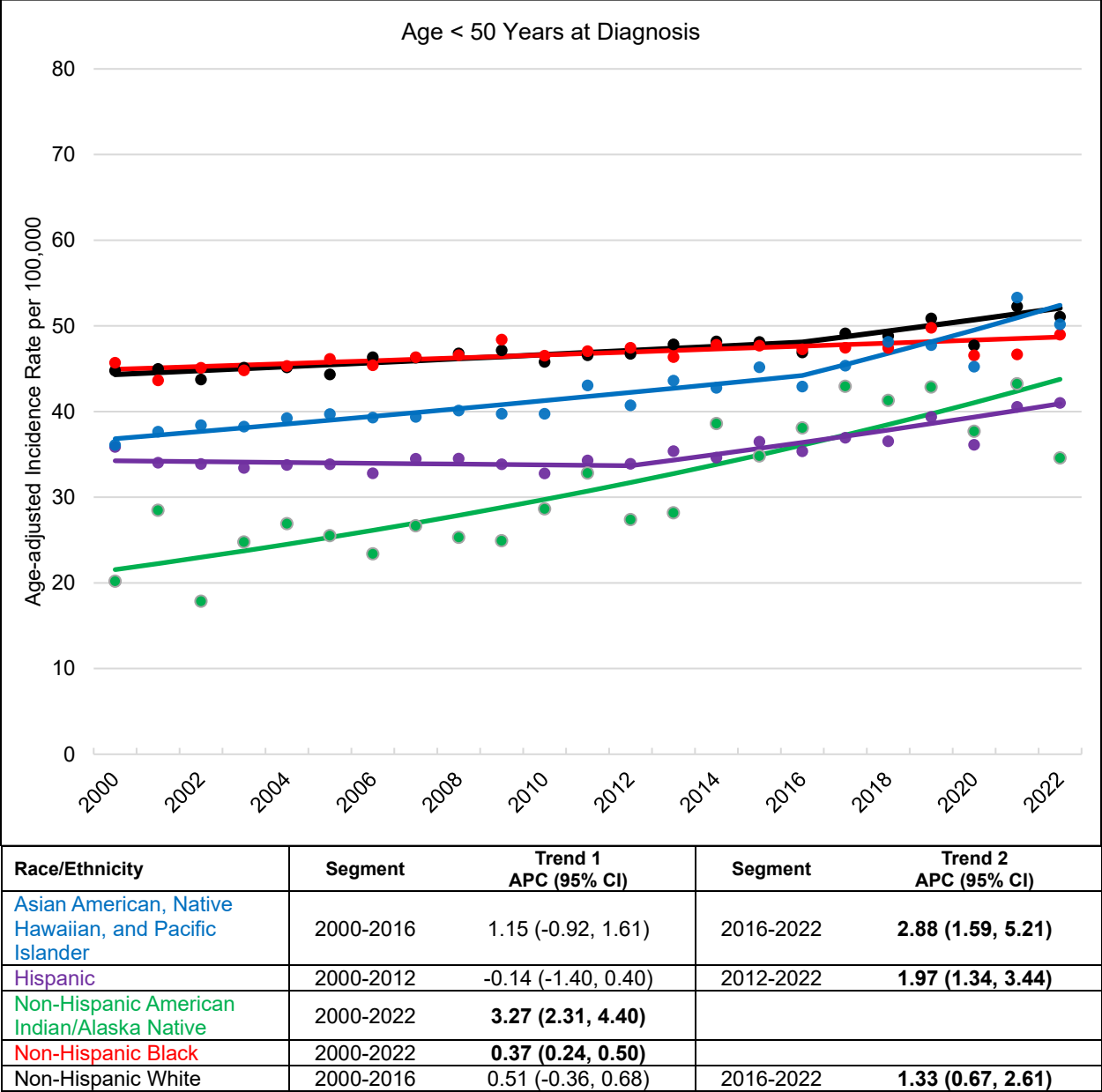
eFigure 1. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, 2000-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.

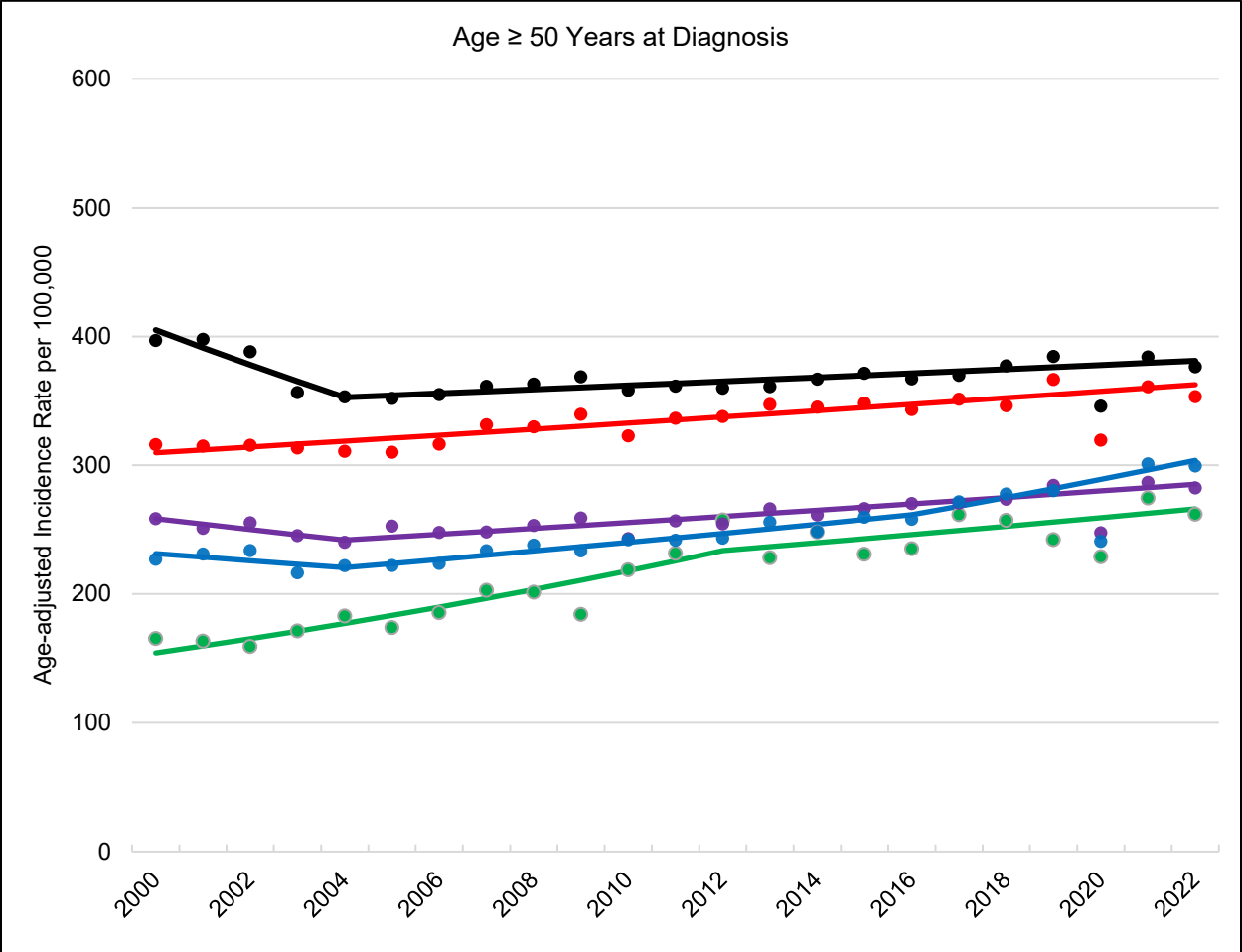
eFigure 2. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, 2000-2022, SEER-21

A. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, age < 50 years at diagnosis, 2000-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.

B. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, age ≥ 50 years at diagnosis, 2000-2022, SEER-21

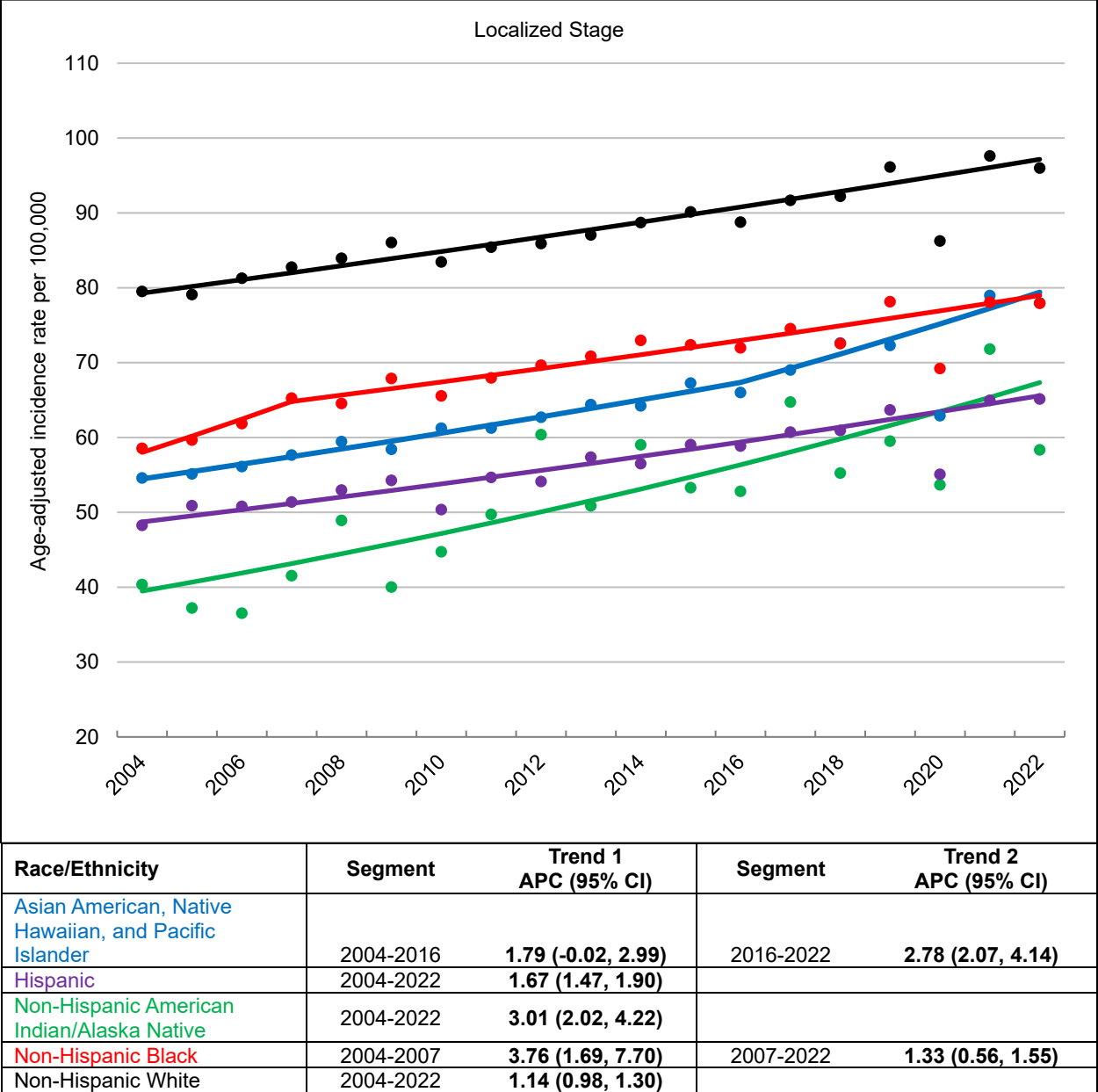


Race/Ethnicity	Segment	Trend 1 APC (95% CI)	Segment	Trend 2 APC (95% CI)	Segment	Trend 3 APC (95% CI)
Asian American, Native Hawaiian, and Pacific Islander	2000-2004	-1.20 (-4.57, 1.54)	2004-2016	1.44 (0.12, 3.46)	2016-2022	2.52 (1.62, 3.91)
Hispanic	2000-2004	-1.64 (-5.05, 0.23)	2004-2022	0.92 (0.76, 1.23)		
Non-Hispanic American Indian/Alaska Native	2000-2012	3.53 (2.68, 12.01)	2012-2022	1.31 (-1.39, 2.29)		
Non-Hispanic Black	2000-2022	0.72 (0.58, 0.89)				
Non-Hispanic White	2000-2004	-3.40 (-6.95, -1.73)	2004-2022	0.43 (0.25, 0.67)		

Bold text indicates a statistically significant increase/decrease in trend.

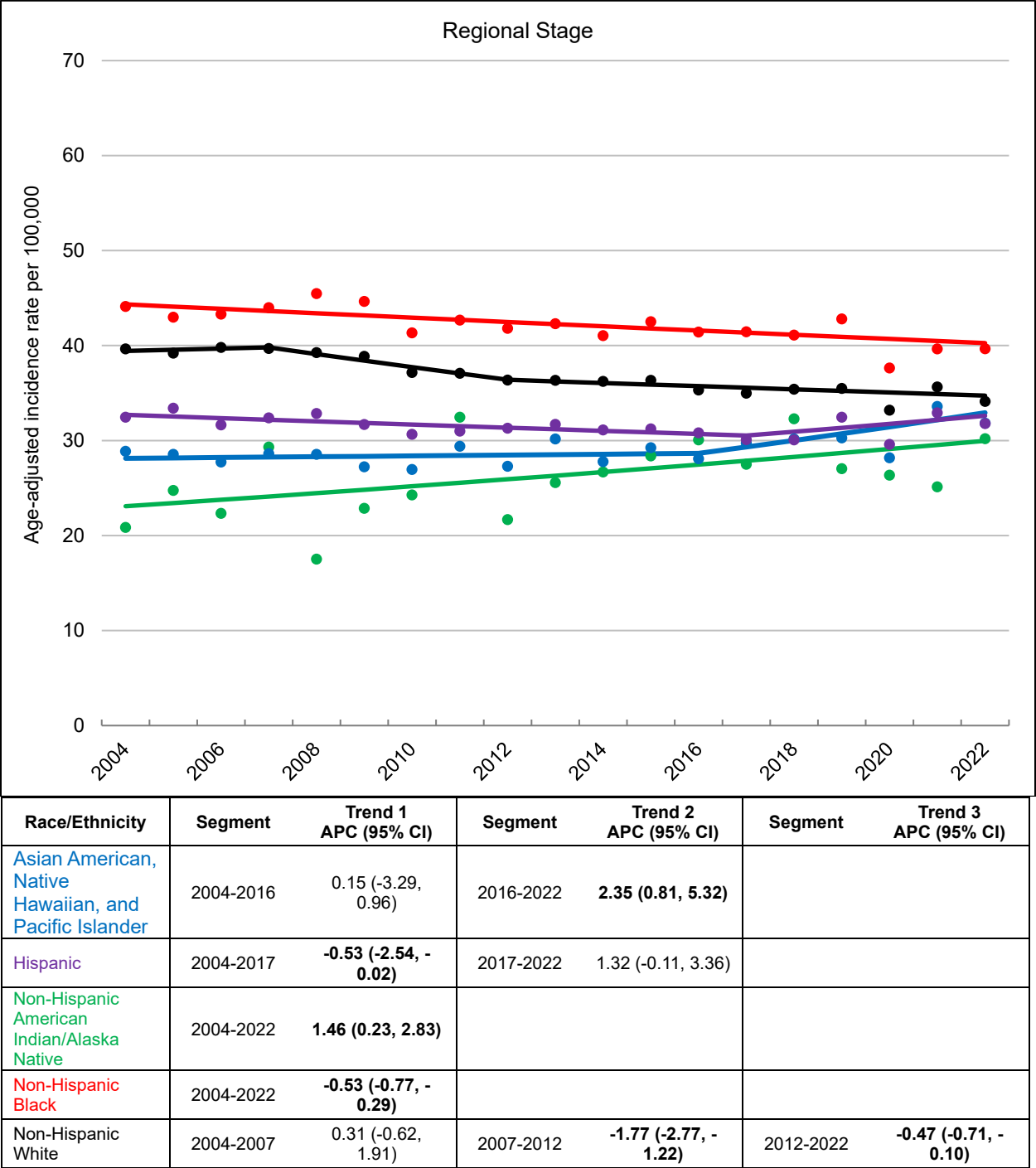
eFigure 3. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, by Stage, 2004-2022, SEER-21

A. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, localized stage, 2004-2022, SEER-21



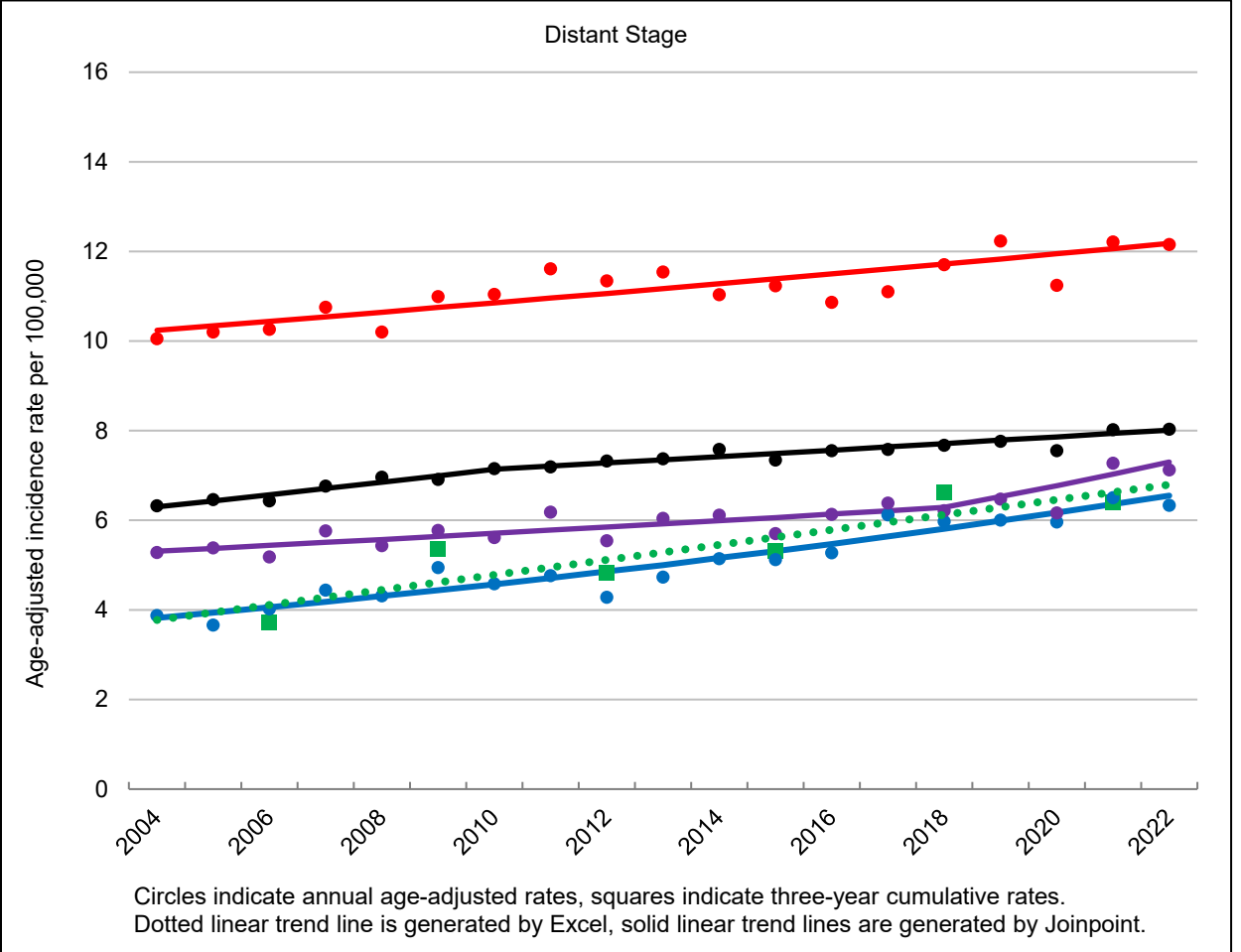
Bold text indicates a statistically significant increase/decrease in trend.

B. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, regional stage, 2004-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.

C. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, distant stage, 2004-2022, SEER-21

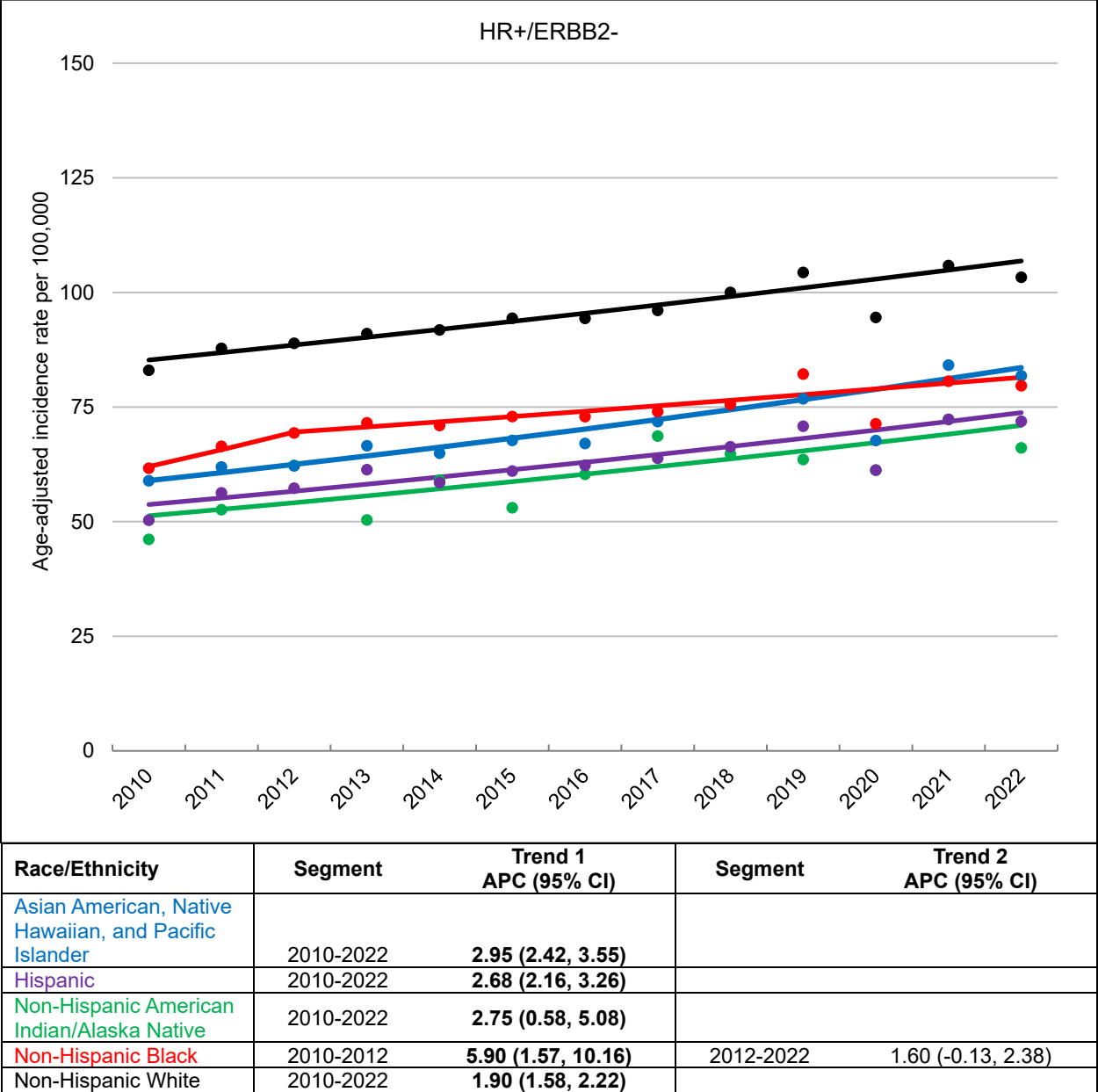


Race/Ethnicity	Segment	Trend 1 APC (95% CI)	Segment	Trend 2 APC (95% CI)
Asian American, Native Hawaiian, and Pacific Islander	2004-2022	3.04 (2.51, 3.71)		
Hispanic	2004-2018	1.22 (-2.49, 5.62)	2018-2022	3.78 (1.48, 6.60)
Non-Hispanic American Indian/Alaska Native	2005-2022	3.03 (1.28, 5.11) ^		
Non-Hispanic Black	2004-2022	0.97 (0.65, 1.33)		
Non-Hispanic White	2004-2010	2.10 (1.62, 3.17)	2010-2022	0.97 (0.70, 1.14)

^ Tri-annual percentage change: 2005-2007, 2008-2010, 2011-2013, 2014-2016, 2017-2019, 2020-2022 * excl 2004
Bold text indicates a statistically significant increase/decrease in trend.

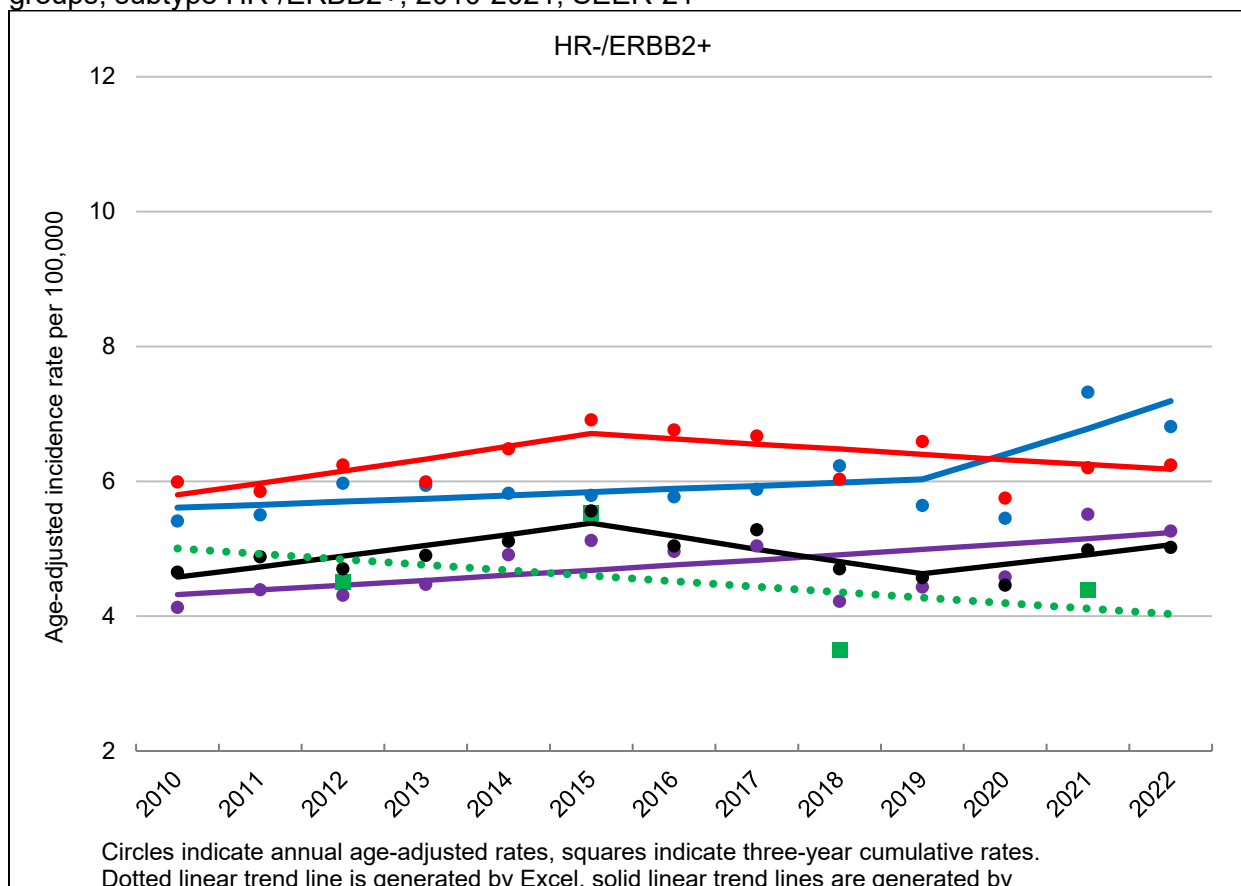
eFigure 4. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, by Subtype, 2010-2022, SEER-21

A. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, subtype HR+/ERBB2-, 2010-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.

B. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, subtype HR-/ERBB2+, 2010-2021, SEER-21

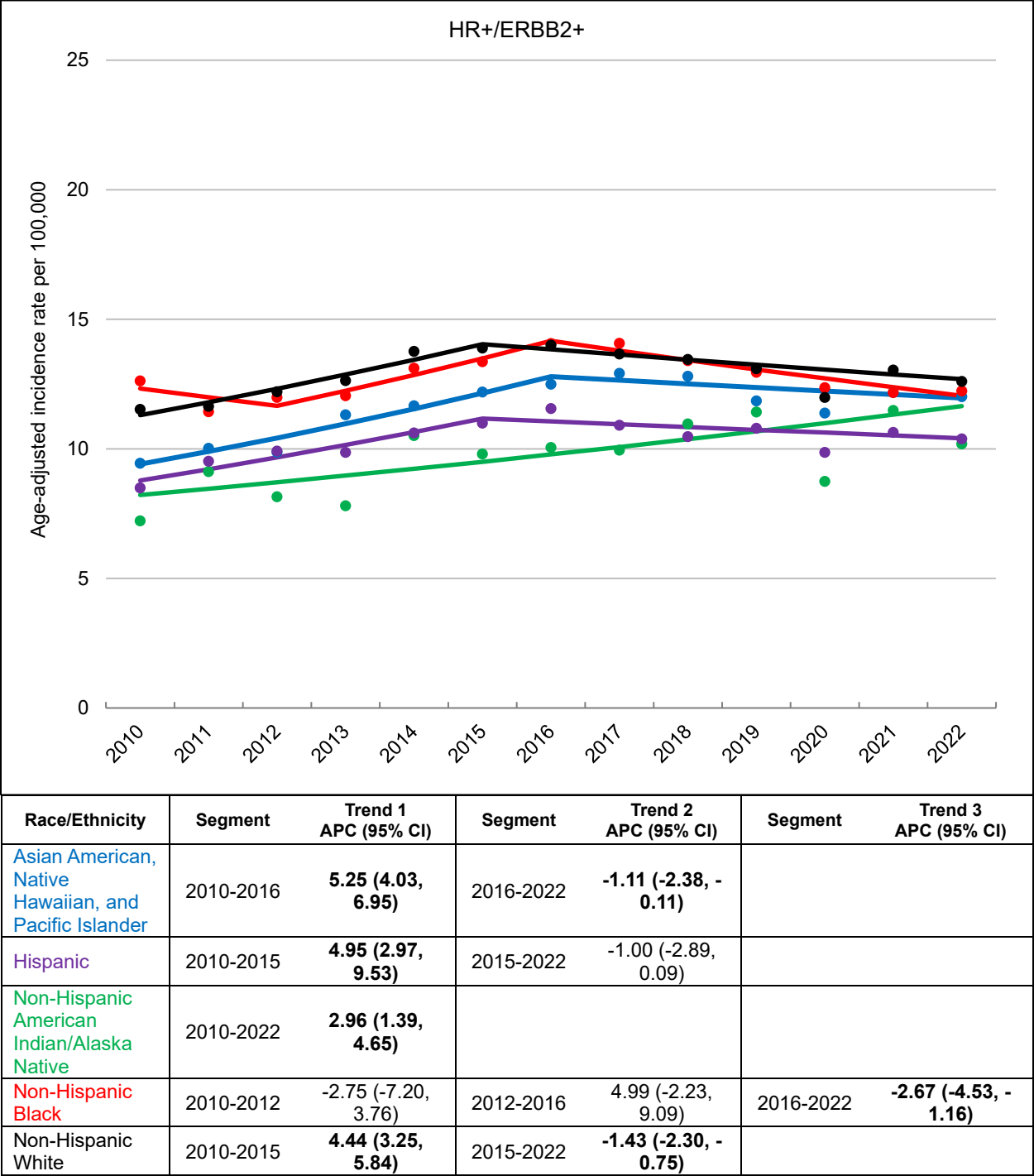


Race/Ethnicity	Segment	Trend 1 APC (95% CI)	Segment	Trend 2 APC (95% CI)	Segment	Trend 3 APC (95% CI)
Asian American, Native Hawaiian, and Pacific Islander	2010-2019	0.81 (-5.81, 10.00)	2019-2022	6.05 (0.97, 10.21)		
Hispanic	2010-2022	1.62 (0.23, 3.11)				
Non-Hispanic American Indian/Alaska Native	2011-2022	-1.87 (-12.38, 9.63) ^				
Non-Hispanic Black	2010-2015	2.97 (1.23, 7.83)	2015-2022	-1.18 (-3.28, -0.15)		
Non-Hispanic White	2010-2015	3.30 (0.71, 9.02)	2015-2019	-3.70 (-8.60, 3.50)	2019-2022	3.03 (-2.02, 7.94)

^ Tri-annual percentage change: 2011-2013, 2014-2016, 2017-2019, 2020-2022 *excl 2010

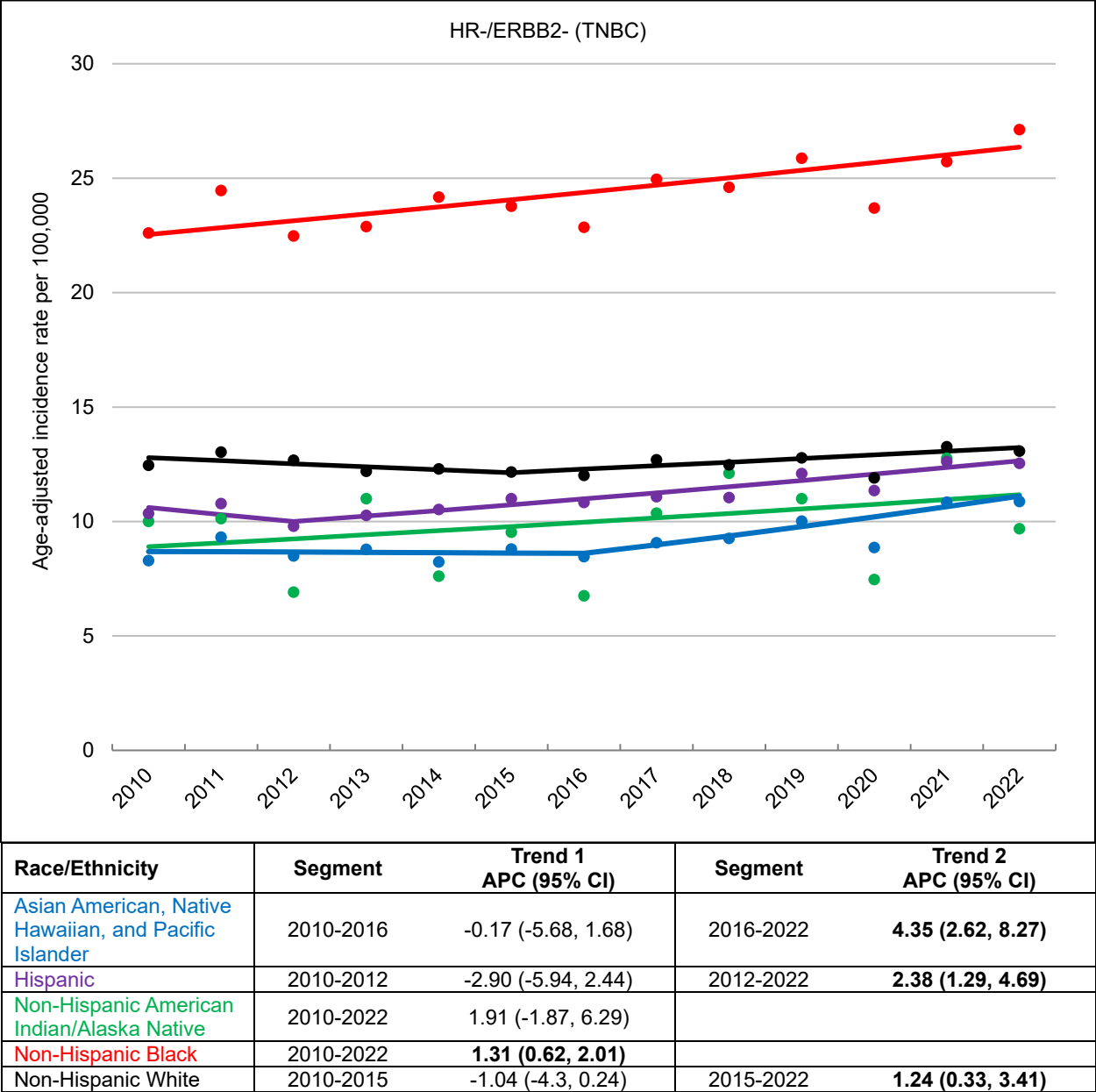
Bold text indicates a statistically significant increase/decrease in trend.

C. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, subtype HR+/ERBB2+, 2010-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.

D. Age-adjusted invasive breast cancer incidence rates and Joinpoint trends by aggregated racial/ethnic groups, subtype HR-/ERBB2- (TNBC), 2010-2022, SEER-21



Bold text indicates a statistically significant increase/decrease in trend.